

Initiating Search

November 10, 2025, 10:17 PM

• Search:

CAS Lexicon Search:

Kabachnik-Fields condensation reaction - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (510)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (619)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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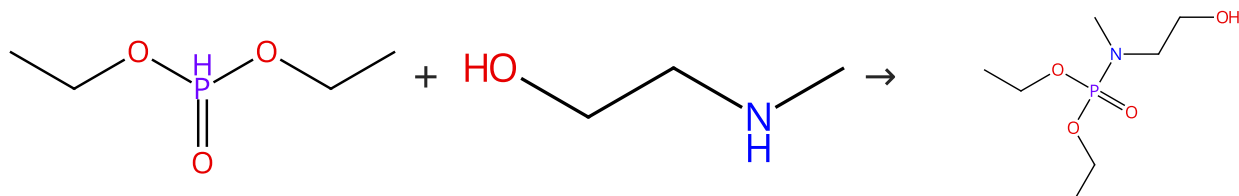
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (46)

 Suppliers (75)

31-614-CAS-28450337

Steps: 1 Yield: 100%

Microwave-assisted synthesis of α -aminophosphonates and related derivatives by the Kabachnik-Fields reaction

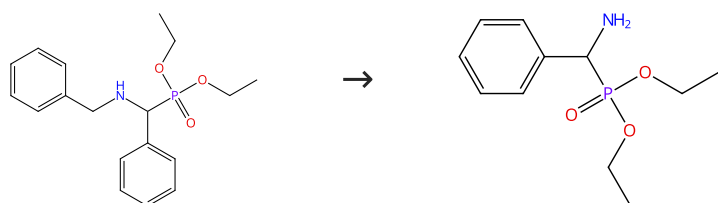
By: Tajti, Adam; et al

Phosphorus, Sulfur and Silicon and the Related Elements (2019), 194(4-6), 379-381.

1.1 Reagents: Formaldehyde; 20 min, 100 °C

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (4)

 Suppliers (8)

31-614-CAS-37310860

Steps: 1 Yield: 100%

New N-acyl- as well as N-phosphonoylmethyl- and N-phosphinoylmethyl- α -amino-benzylphosphonates by acylation and a tandem Kabachnik-Fields protocol

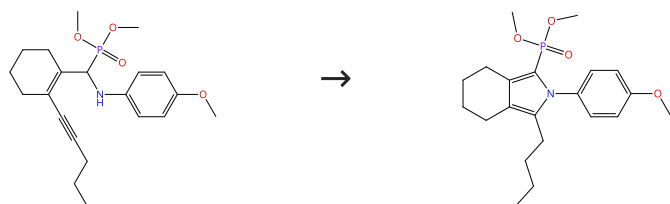
By: Varga, Petra Regina; et al

Organic & Biomolecular Chemistry (2023), 21(8), 1709-1718.

1.1 Reagents: Hydrogen
Catalysts: Palladium
Solvents: Ethanol; 24 h, 12 bar, 80 °C

Scheme 3 (1 Reaction)

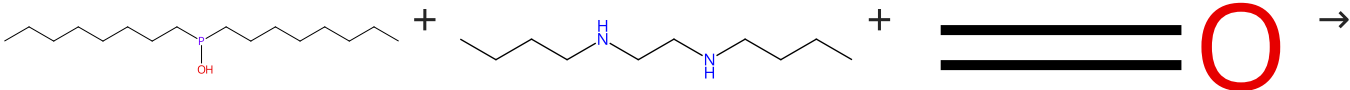
Steps: 1 Yield: 99%



31-614-CAS-27586476	Steps: 1 Yield: 99%	Formation of Condensed 1 H-Pyrrol-2-ylphosphonates and 1, 2-Dihydropyridin-2-ylphosphonates via Kabachnik-Fields Reaction of Acetylenic Aldehydes and Subsequent 5-exo-dig or 6-endo-dig Cyclizations
1.1 Reagents: Potassium <i>tert</i> -butoxide Catalysts: Gold tribromide Solvents: Chloroform; 0.5 - 2 h, rt		By: Buksnaitiene, Rita; et al Journal of Organic Chemistry (2014), 79(14), 6532-6553.
Experimental Protocols		

Scheme 4 (1 Reaction)

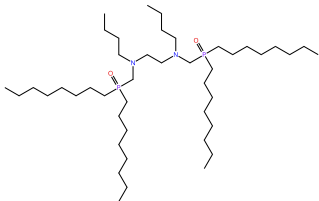
Steps: 1 Yield: 99%



 Suppliers (2)

 Suppliers (19)

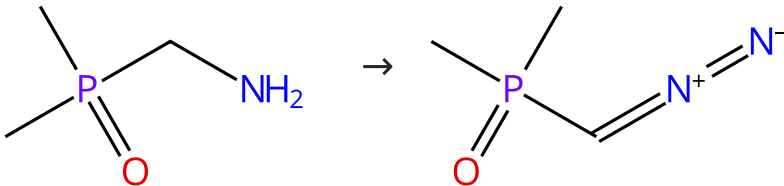
 Suppliers (188)



31-614-CAS-29910193	Steps: 1 Yield: 99%	Synthesis and ionophoric properties of α, ω -diphosphorylated aza podands: II. Kabachnik-fields reaction as a method for design of α, ω -diphosphorylated aza podands and their use for determination of metal ions
1.1 Solvents: Toluene; 3 h, reflux		By: Garifzyanov, A. R.; et al Russian Journal of General Chemistry (2005), 75(2), 217-224.
Experimental Protocols		

Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%



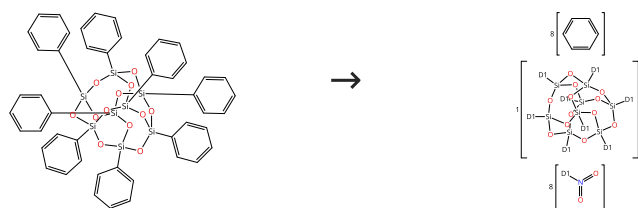
The reaction scheme shows the conversion of dimethylphosphinic acid to a diazo compound. The reactant is dimethylphosphinic acid, represented by a phosphorus atom (P) double-bonded to an oxygen atom (O) and single-bonded to two methyl groups and a methanamine group (-CH₂NH₂). The product is a diazo compound, represented by a phosphorus atom (P) double-bonded to an oxygen atom (O) and single-bonded to two methyl groups and a diazomethyl group (-CH₂N⁺=N⁻). The reaction is indicated by a right-pointing arrow.

 Suppliers (6)

31-614-CAS-41862572	Steps: 1 Yield: 99%	(Diazomethyl)dimethylphosphine Oxide - A Diazoalkane Reagent for [3+2] Cycloadditions
1.1 Reagents: <i>tert</i> -Butyl nitrite Catalysts: Acetic acid Solvents: Chloroform; 10 min, rt; 45 min, 65 °C		By: Slobodyanyuk, Evgeniy Y.; et al Chemistry - A European Journal (2024), 30(23), e202303972.
Experimental Protocols		

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (49)

31-614-CAS-44586926

Steps: 1 Yield: 99%

Mechanically reinforced and flame-retardant epoxy resin nanocomposite based on molecular engineering of POSS

 1.1 **Reagents:** Sulfuric acid, Nitric acid; 30 min, 0 °C; 0 °C → rt; 20 h, rt

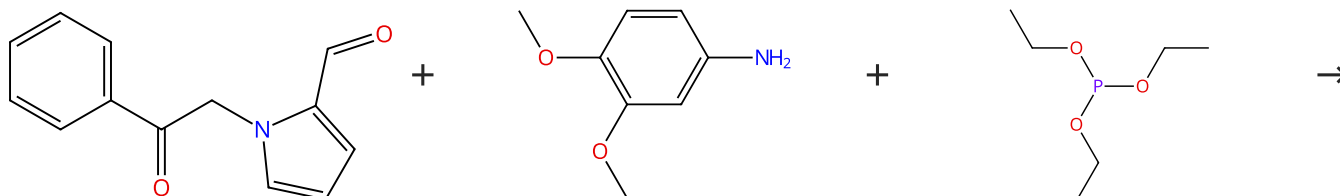
By: Liu, Cheng; et al

Experimental Protocols

Polymer Testing (2025), 143, 108719.

Scheme 7 (1 Reaction)

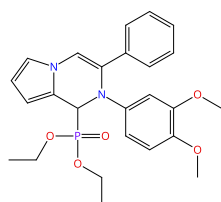
Steps: 1 Yield: 98%



Suppliers (7)

Suppliers (95)

Suppliers (43)



31-614-CAS-28635892

Steps: 1 Yield: 98%

One-pot three-component coupling access to 1,2-dihydro pyrrolo[1,2-a]pyrazine-1-phosphonates: multi-functionalization of a pyrazine unit

 1.1 **Catalysts:** Scandium triflate
Solvents: Ethanol; 24 h, rt

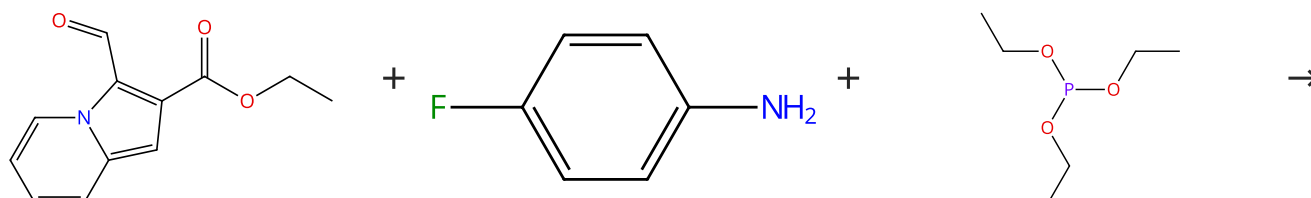
By: Lee, Jeong Hwa; et al

Experimental Protocols

Organic & Biomolecular Chemistry (2021), 19(27), 6066-6084.

Scheme 8 (1 Reaction)

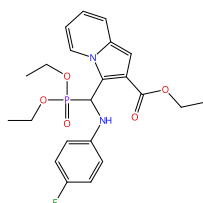
Steps: 1 Yield: 98%



Suppliers (2)

Suppliers (70)

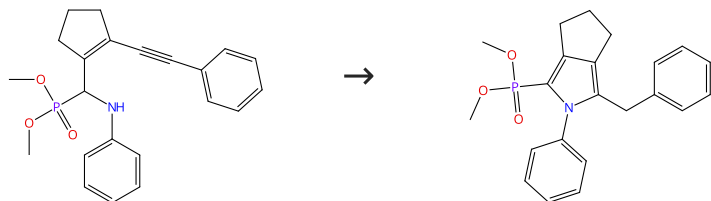
Suppliers (43)



31-614-CAS-26129579 Steps: 1 Yield: 98% 1.1 Catalysts: Iron chloride (FeCl₃) Solvents: Ethanol; 24 h, 40 °C	Diversity-oriented functionalization of indolizines at the C3 position via multicomponent Kabachnik-Fields reaction By: Kim, Jinwoo; et al Tetrahedron (2017), 73(39), 5759-5768.
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Scheme 9 (1 Reaction)

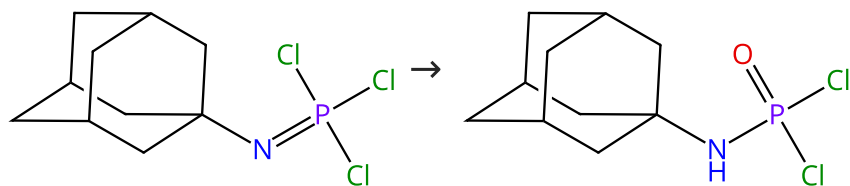
Steps: 1 Yield: 98%



31-614-CAS-25065918 Steps: 1 Yield: 98% 1.1 Reagents: Potassium <i>tert</i>-butoxide Catalysts: Gold tribromide Solvents: Chloroform; 0.5 - 2 h, rt Experimental Protocols	Formation of Condensed 1 H-Pyrrol-2-ylphosphonates and 1, 2-Dihydropyridin-2-ylphosphonates via Kabachnik-Fields Reaction of Acetylenic Aldehydes and Subsequent 5-exo-dig or 6-endo-dig Cyclizations By: Buksnaitiene, Rita; et al Journal of Organic Chemistry (2014), 79(14), 6532-6553.
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Scheme 10 (1 Reaction)

Steps: 1 Yield: 98%



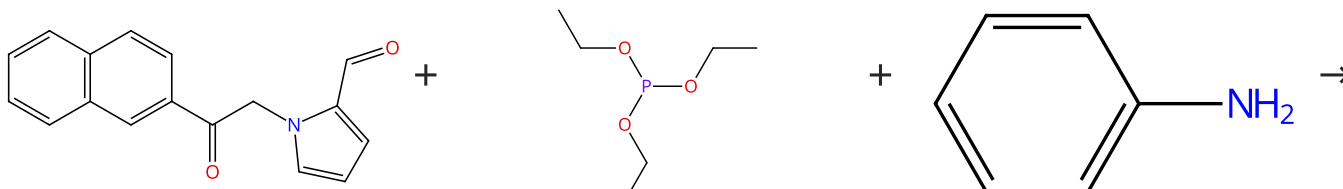
Supplier (1)

Suppliers (5)

31-614-CAS-30546320 Steps: 1 Yield: 98% 1.1 Reagents: Water Solvents: Diethyl ether	Phosphorylated adamantanes. XVII. Phosphorylation of adamantane amines By: Yurchenko, R. I.; et al Zhurnal Obshchei Khimii (1995), 65(9), 1449-53.
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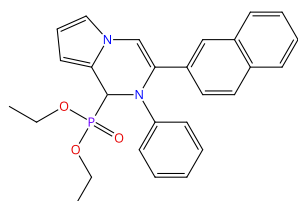
Scheme 11 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (43)

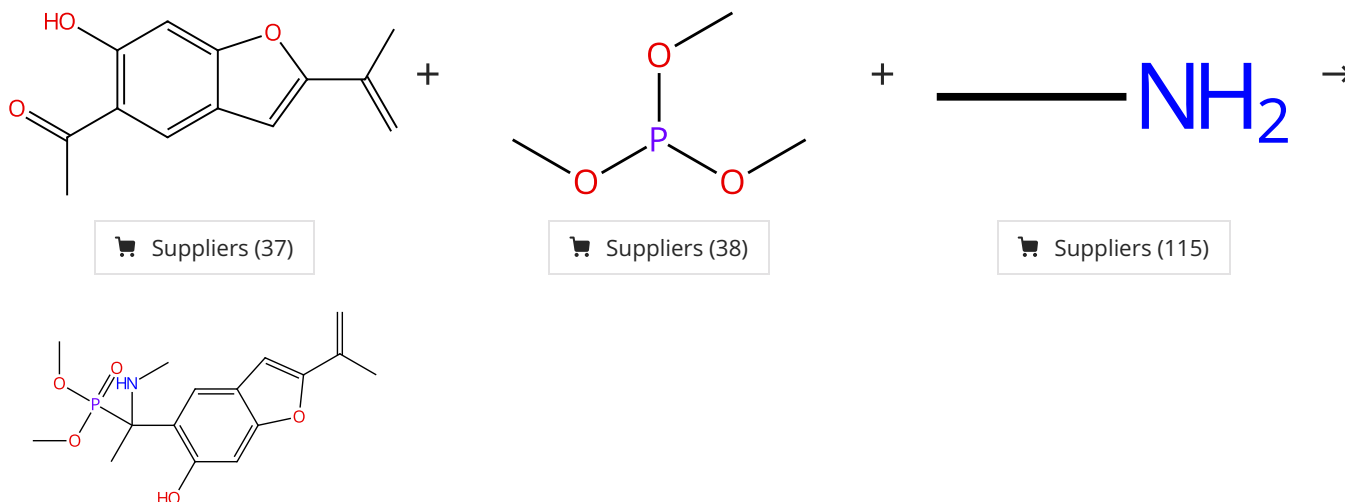
Suppliers (113)



31-614-CAS-27092183	Steps: 1 Yield: 98%	One-pot three-component coupling access to 1,2-dihydro pyrrolo[1,2-a]pyrazine-1-phosphonates: multi-functiona lization of a pyrazine unit
1.1 Catalysts: Scandium triflate Solvents: Ethanol; 24 h, rt		By: Lee, Jeong Hwa; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2021), 19(27), 6066-6084.

Scheme 12 (1 Reaction)

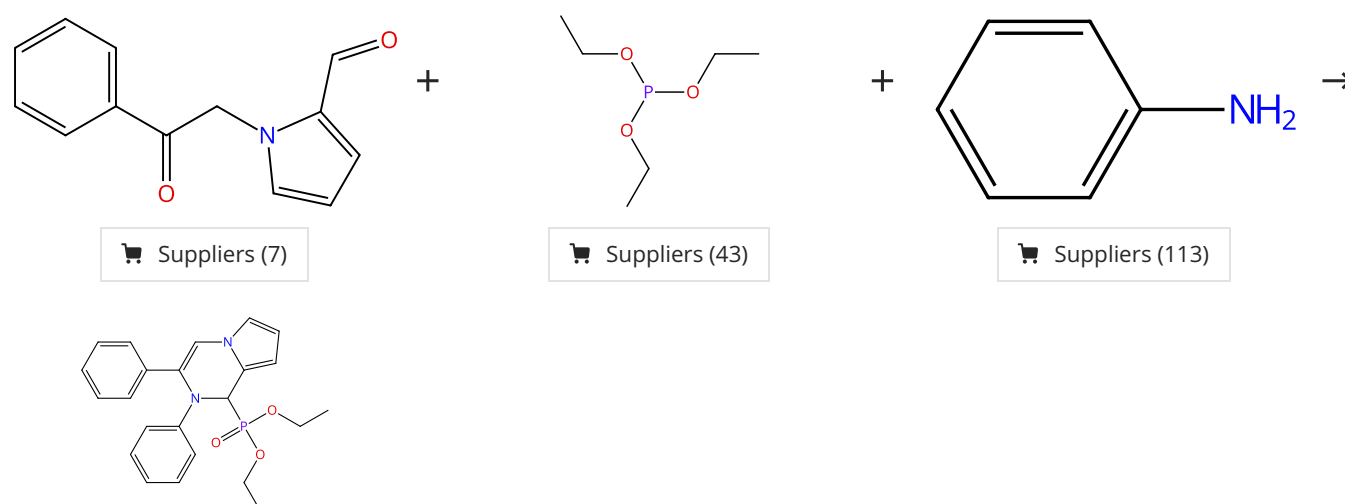
Steps: 1 Yield: 98%



31-614-CAS-27874777	Steps: 1 Yield: 98%	Green synthesis of α-aminophosphonates using ZnO nanoparticles as an efficient catalyst
1.1 Catalysts: Zinc oxide (ZnO) Solvents: Water; 15 min, rt; 1 h, rt		By: Ezzatzadeh, Elham
		Zeitschrift fuer Naturforschung, B: A Journal of Chemical Sciences (2018), 73(3-4), 179-184.

Scheme 13 (1 Reaction)

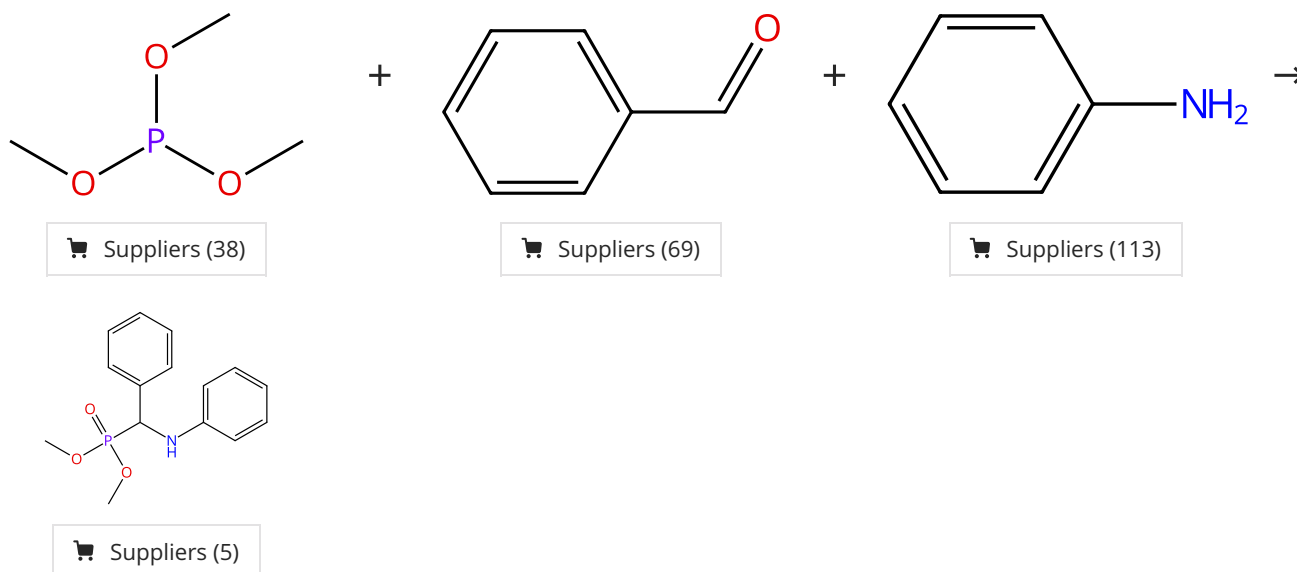
Steps: 1 Yield: 98%



31-614-CAS-25455574	Steps: 1 Yield: 98%	One-pot three-component coupling access to 1,2-dihydro pyrrolo[1,2-a]pyrazine-1-phosphonates: multi-functiona lization of a pyrazine unit
1.1 Catalysts: Scandium triflate Solvents: Ethanol; 24 h, rt		By: Lee, Jeong Hwa; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2021), 19(27), 6066-6084.

Scheme 14 (2 Reactions)

Steps: 1 Yield: 96-98%



31-614-CAS-37833253

Steps: 1 Yield: 98%

1.1 **Catalysts:** Zinc chloride, Poly(vinylpyrrolidone) (quaternized)
Solvents: Ethanol, Water; 15 min, rt

Experimental Protocols

Pvp-based deep eutectic solvent polymer: Sustainable Bronsted-Lewis acidic catalyst in the synthesis of α -aminophosphonate and bisindole

By: Shahiri Haghayegh, Mehran; et al

Journal of Molecular Liquids (2023), 387, 122677.

31-614-CAS-44006376

Steps: 1 Yield: 96%

1.1 **Catalysts:** Lanthanum chloride heptahydrate; 1 h, rt

Experimental Protocols

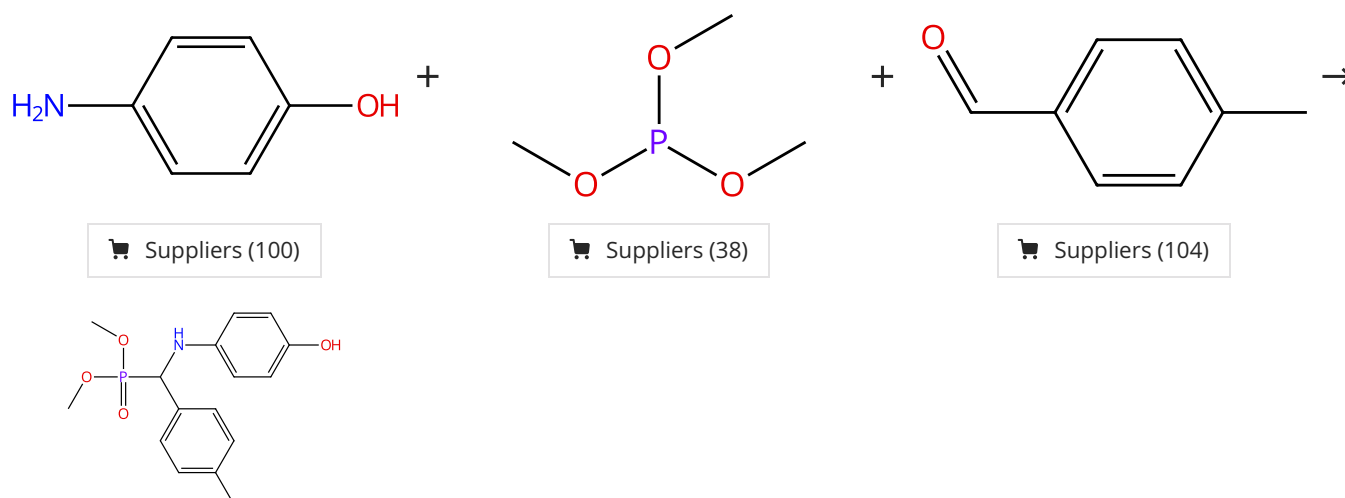
$\text{LaCl}_3 \cdot 7\text{H}_2\text{O}$ as an Effective Catalyst for the Synthesis of α -Aminophosphonates under Solvent-Free Conditions and Docking Simulation of Ligand Bond Complexes of Cyclin-Dependent Kinase 2

By: Mirzaei, Mahdi; et al

Polycyclic Aromatic Compounds (2022), 42(9), 5882-5892.

Scheme 15 (1 Reaction)

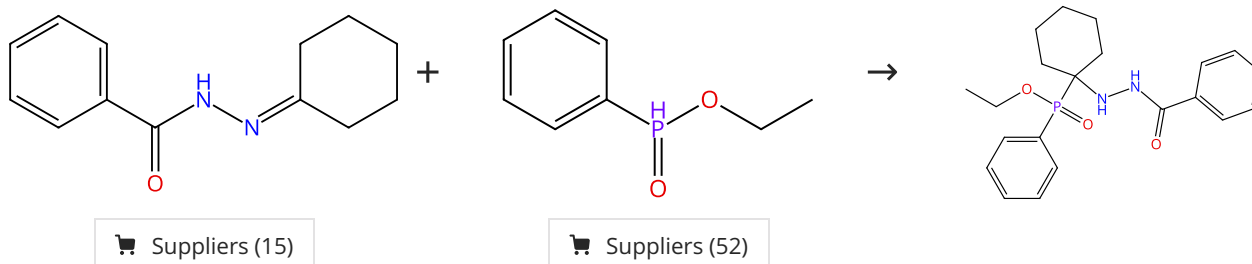
Steps: 1 Yield: 97%



31-614-CAS-38867615 Steps: 1 Yield: 97% 1.1 Solvents: Ethanol; 8 min, reflux Experimental Protocols	Phosphonylated p-aminophenol derivatives: Microwave Assisted Synthesis, Density Functional Theory study and biological activities evaluation By: Kirouani, Imene; et al Journal of Molecular Structure (2023), 1290, 135916.
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Scheme 16 (1 Reaction)

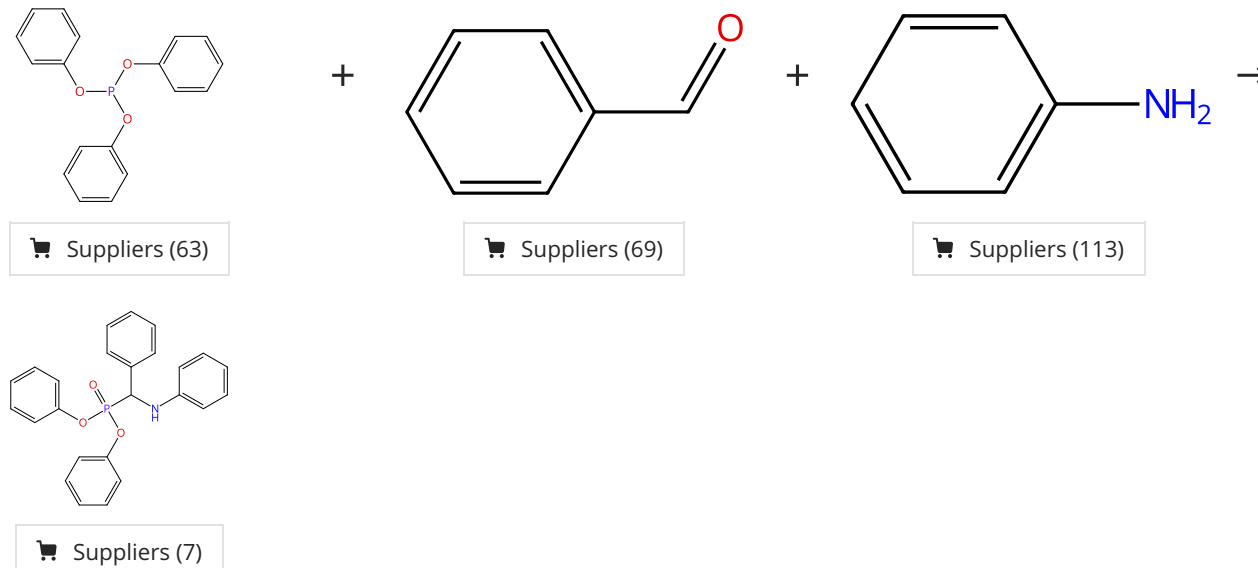
Steps: 1 Yield: 97%



31-614-CAS-29399591 Steps: 1 Yield: 97% 1.1 Catalysts: Chloro[<i>C,C,C,C</i> -tetrakis(1,1-dimethylethyl)-29 <i>H</i> ,31 <i>H</i> -phthalocyaninato(2-)-κ N^{29} ,κ N^{30} ,κ N^{31} ,κ N^{32}]aluminum Solvents: Ethyl phenylphosphinate; 48 h, 80 °C Experimental Protocols	New Possibilities of the Kabachnik-Fields and Pudovik Reactions in the Phthalocyanine-Catalyzed Syntheses of α-Aminophosphonic and α-Aminophosphinic Acid Derivatives By: Shuvalov, M. V.; et al Russian Journal of General Chemistry (2018), 88(9), 1761-1775.
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Scheme 17 (1 Reaction)

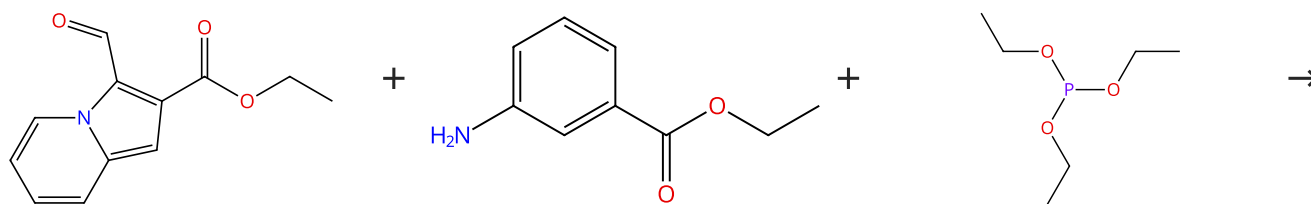
Steps: 1 Yield: 97%



31-614-CAS-37833271 Steps: 1 Yield: 97% 1.1 Catalysts: Zinc chloride, Poly(vinylpyrrolidone) (quaternized) Solvents: Ethanol, Water; 60 min, rt Experimental Protocols	Pvp-based deep eutectic solvent polymer: Sustainable Bronsted-Lewis acidic catalyst in the synthesis of α-aminophosphonate and bisindole By: Shahiri Haghayegh, Mehran; et al Journal of Molecular Liquids (2023), 387, 122677.
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Scheme 18 (1 Reaction)

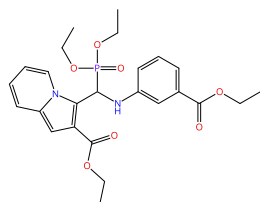
Steps: 1 Yield: 97%



Suppliers (2)

Suppliers (82)

Suppliers (43)



31-614-CAS-30948425

Steps: 1 Yield: 97%

Diversity-oriented functionalization of indolizines at the C3 position via multicomponent Kabachnik-Fields reaction

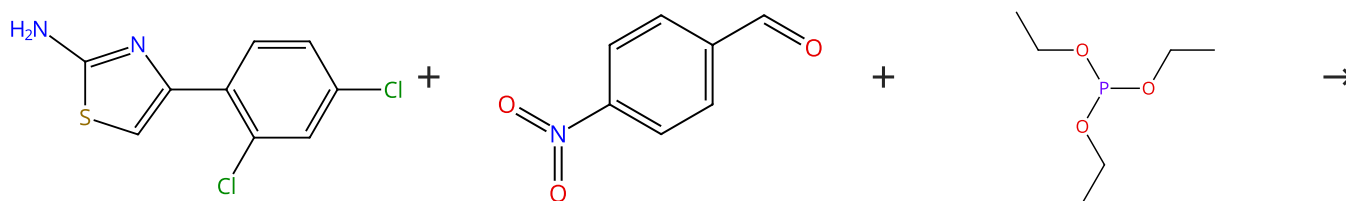
1.1 **Catalysts:** Iron chloride (FeCl₃)
Solvents: Ethanol; 24 h, 40 °C

By: Kim, Jinwoo; et al

Tetrahedron (2017), 73(39), 5759-5768.

Scheme 19 (1 Reaction)

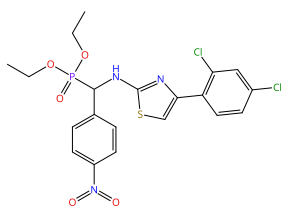
Steps: 1 Yield: 97%



Suppliers (64)

Suppliers (92)

Suppliers (43)



31-614-CAS-27124168

Steps: 1 Yield: 97%

Solvent-free sonochemical Kabachnik-Fields reaction to synthesize some new α-aminophosphonates catalyzed by nano-BF₃•SiO₂

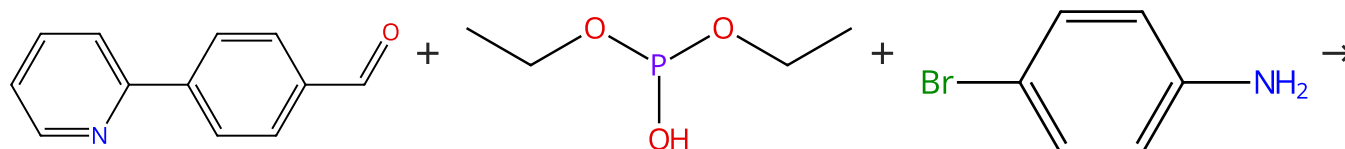
1.1 **Catalysts:** Silica, Boron trifluoride (silica supported); 12 h, rt
 Experimental Protocols

By: Ravikumar, D.; et al

Phosphorus, Sulfur and Silicon and the Related Elements (2018), 193(6), 400-407.

Scheme 20 (1 Reaction)

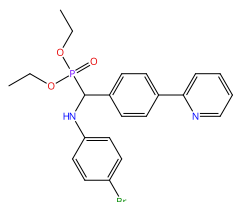
Steps: 1 Yield: 97%



Suppliers (99)

Suppliers (10)

Suppliers (83)



31-614-CAS-25064289

Steps: 1 Yield: 97%

Design, synthesis, antioxidant, and anti-breast cancer activities of novel diethyl(alkyl/aryl/heteroaryl amino)(4-(pyridin-2-yl)phenyl)methylphosphonates

By: Prasad, Sthanikam Siva; et al

Archiv der Pharmazie (Weinheim, Germany) (2013), 346(5), 380-391.

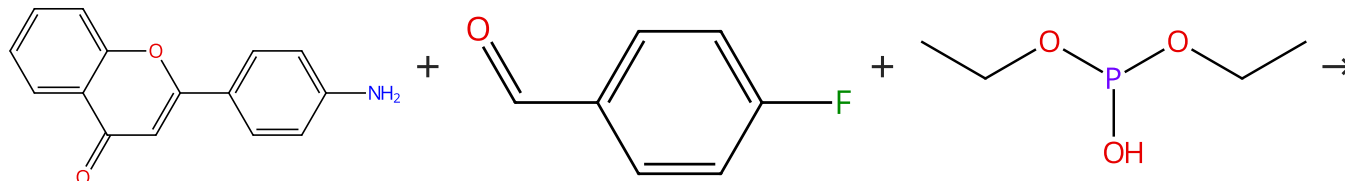
1.1 Catalysts: Cupric acetate; 4 min

1.2 Reagents: Water

Experimental Protocols

Scheme 21 (1 Reaction)

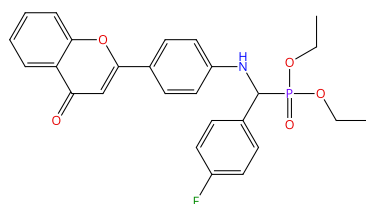
Steps: 1 Yield: 97%



Suppliers (5)

Suppliers (96)

Suppliers (10)



31-614-CAS-38142906

Steps: 1 Yield: 97%

Ultrasound mediated nano ZnO catalyzed synthesis of new α -aminophosphonates as potential anti-diabetic agents; an in silico ADMET, molecular docking study, α -amylase and α -glucosidase inhibitory activity

By: Subramanyam, CH.; et al

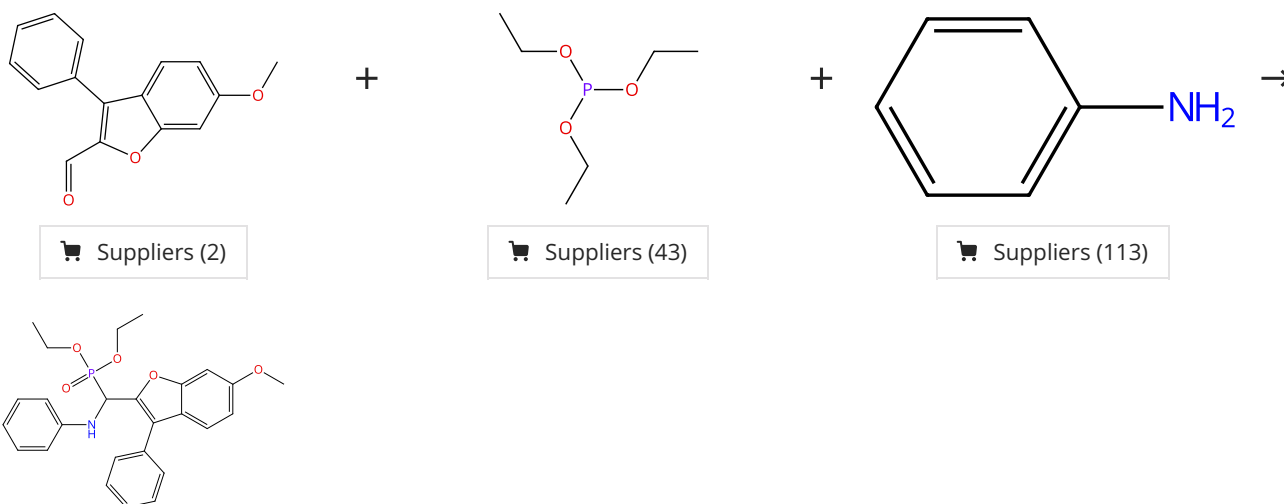
Synthetic Communications (2023), 53(23), 2041-2060.

1.1 Catalysts: Zinc oxide (ZnO); 10 min, rt

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-27112665

Steps: 1 Yield: 96%

Expanding the chemical space: Discovery of new anticancer 3-arylbenzofuran derivatives

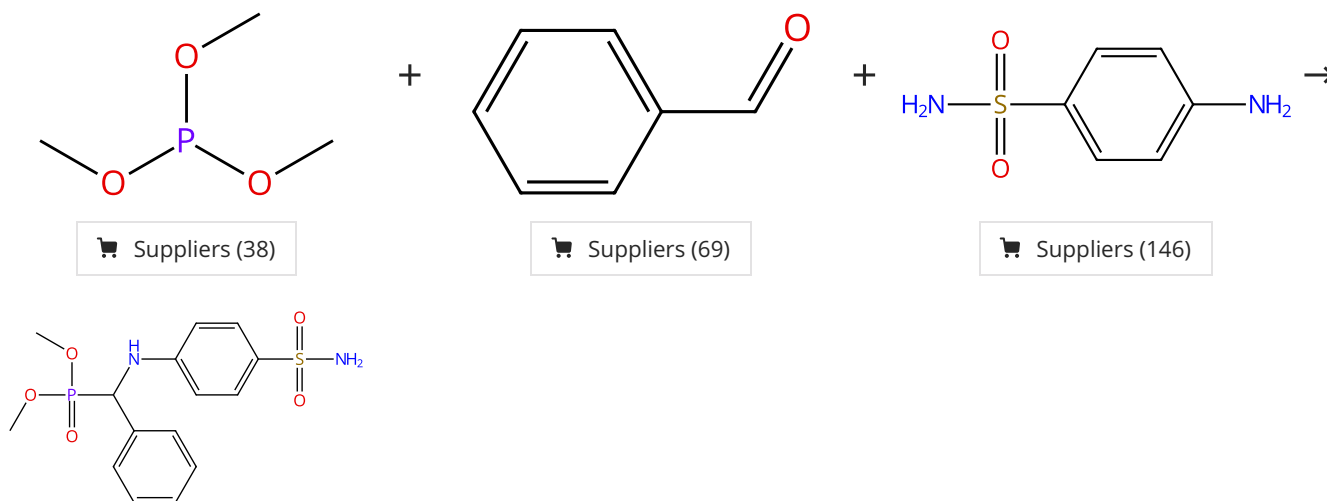
By: Kim, Jinhwang; et al

Journal of Heterocyclic Chemistry (2020), 57(9), 3279-3293.

1.1 Catalysts: Iron chloride (FeCl₃)
Solvents: Ethanol; 16 h, 100 °C

Scheme 23 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-34850479

Steps: 1 Yield: 96%

Sulfonamide-phosphonate hybrids as new carbonic anhydrase inhibitors: In vitro enzymatic inhibition, molecular modeling, and ADMET prediction

By: Zareei, Samira; et al

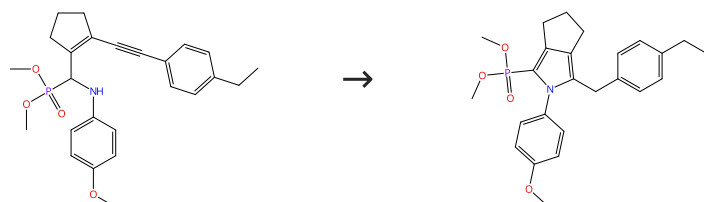
Journal of Molecular Structure (2023), 1271, 134114.

1.1 Solvents: Methanol; 3 h, 80 °C

Experimental Protocols

Scheme 24 (1 Reaction)

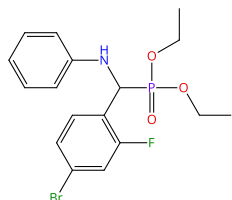
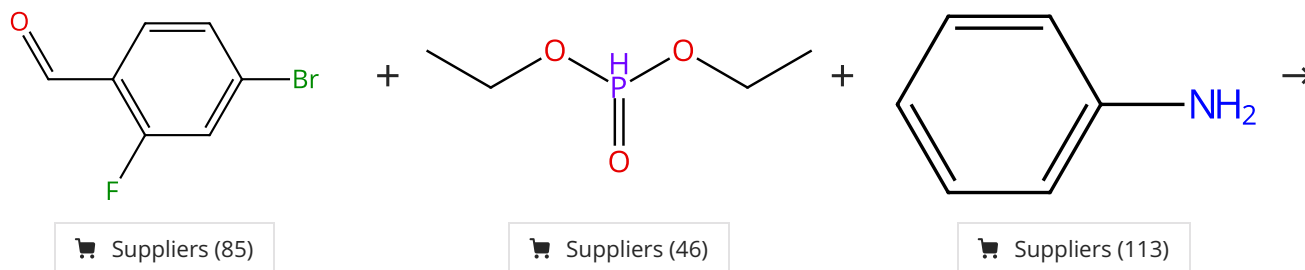
Steps: 1 Yield: 96%



31-614-CAS-27953533	Steps: 1 Yield: 96%	Formation of Condensed 1 H-Pyrrol-2-ylphosphonates and 1, 2-Dihydropyridin-2-ylphosphonates via Kabachnik-Fields Reaction of Acetylenic Aldehydes and Subsequent 5-exo-dig or 6-endo-dig Cyclizations
1.1 Reagents: Potassium <i>tert</i> -butoxide Catalysts: Gold tribromide Solvents: Chloroform; 0.5 - 2 h, rt		By: Buksnaitiene, Rita; et al
Experimental Protocols		Journal of Organic Chemistry (2014), 79(14), 6532-6553.

Scheme 25 (1 Reaction)

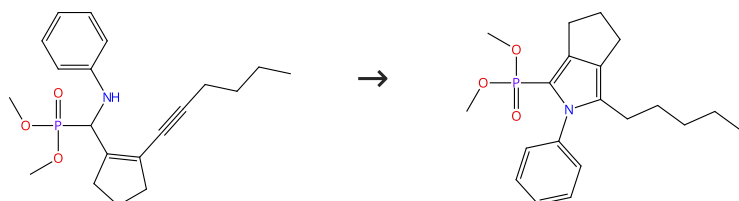
Steps: 1 Yield: 96%



31-614-CAS-47217980	Steps: 1 Yield: 96%	N-(1-hydroxybutan-2-yl)-4-nitrobenzene sulfonamide: A novel organocatalyst for an effective one-pot synthesis of α -amino phosphonates
1.1 Catalysts: <i>N</i> -[1-(Hydroxymethyl)propyl]-4-nitrobenzenesulfonamide Solvents: Ethanol; 15 min, 90 °C; 90 °C → rt		By: Jaydeokar, Swati; et al
1.2 Reagents: Water; rt		Results in Chemistry (2025), 16, 102295.
Experimental Protocols		

Scheme 26 (1 Reaction)

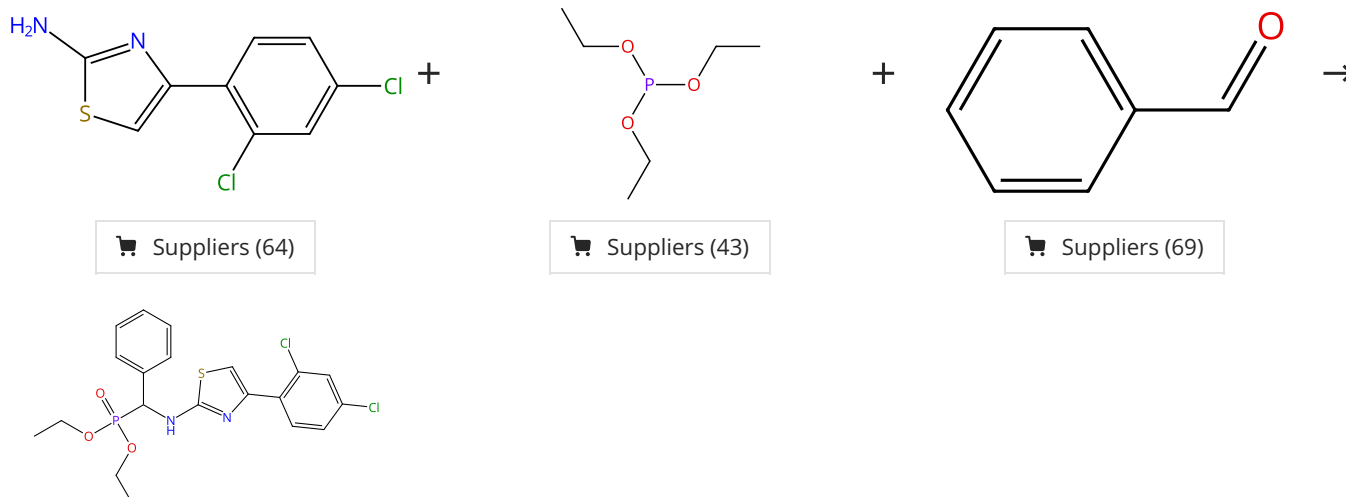
Steps: 1 Yield: 96%



31-614-CAS-28743743	Steps: 1 Yield: 96%	Formation of Condensed 1 H-Pyrrol-2-ylphosphonates and 1, 2-Dihydropyridin-2-ylphosphonates via Kabachnik-Fields Reaction of Acetylenic Aldehydes and Subsequent 5-exo-dig or 6-endo-dig Cyclizations
1.1 Reagents: Potassium <i>tert</i> -butoxide Catalysts: Gold tribromide Solvents: Chloroform; 0.5 - 2 h, rt		By: Buksnaitiene, Rita; et al
Experimental Protocols		Journal of Organic Chemistry (2014), 79(14), 6532-6553.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-25504156

Steps: 1 Yield: 96%

Solvent-free sonochemical Kabachnik-Fields reaction to synthesize some new α -aminophosphonates catalyzed by nano-BF₃·SiO₂

By: Ravikumar, D.; et al

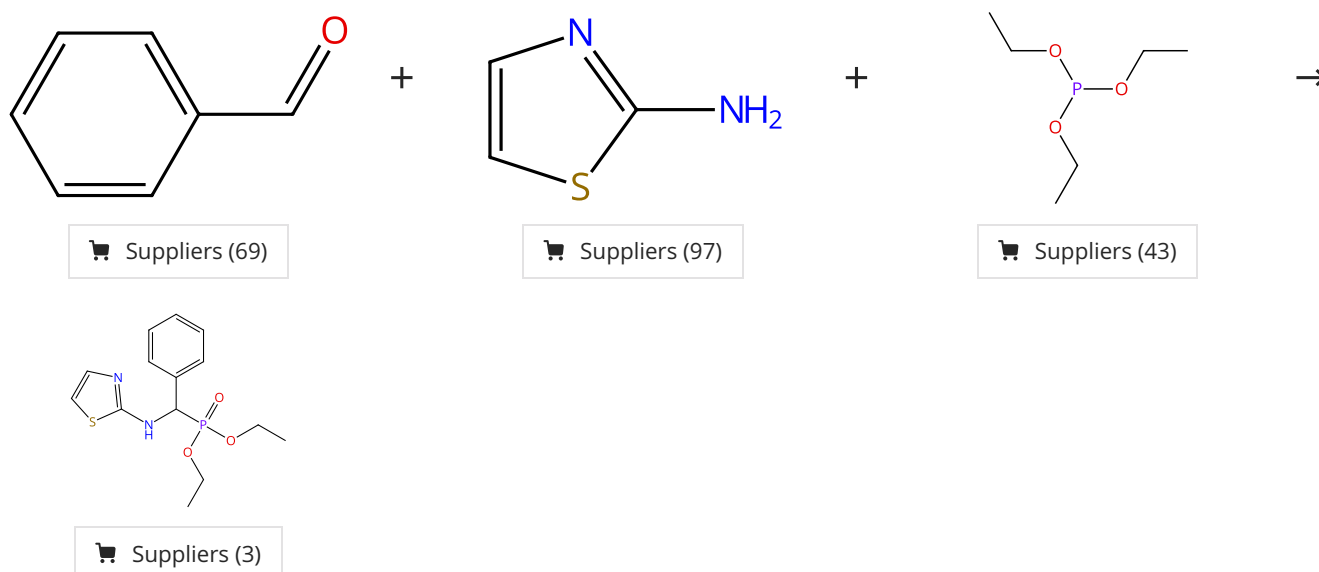
Phosphorus, Sulfur and Silicon and the Related Elements (2018), 193(6), 400-407.

1.1 Catalysts: Silica, Boron trifluoride (silica supported); 10 min, rt

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-28746676

Steps: 1 Yield: 96%

H₆P₂W₁₈O₆₂·14H₂O Catalyzed Synthesis, Spectral Characterization and X-ray Study of α -Aminophosphonates Containing Amino-thiazole Moiety

By: Boughaba, Sara; et al

Current Organic Chemistry (2018), 22(13), 1335-1341.

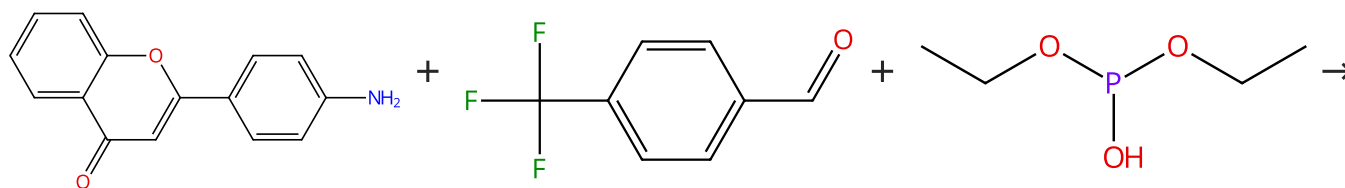
1.1 Catalysts: Tungstophosphoric acid (H₆P₂W₁₈O₆₂); rt

1.2 rt

Experimental Protocols

Scheme 29 (1 Reaction)

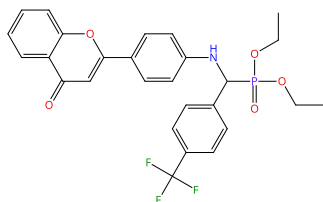
Steps: 1 Yield: 96%



Suppliers (5)

Suppliers (84)

Suppliers (10)



31-614-CAS-38142908

Steps: 1 Yield: 96%

Ultrasound mediated nano ZnO catalyzed synthesis of new α -aminophosphonates as potential anti-diabetic agents; an in silico ADMET, molecular docking study, α -amylase and α -glucosidase inhibitory activity

By: Subramanyam, CH.; et al

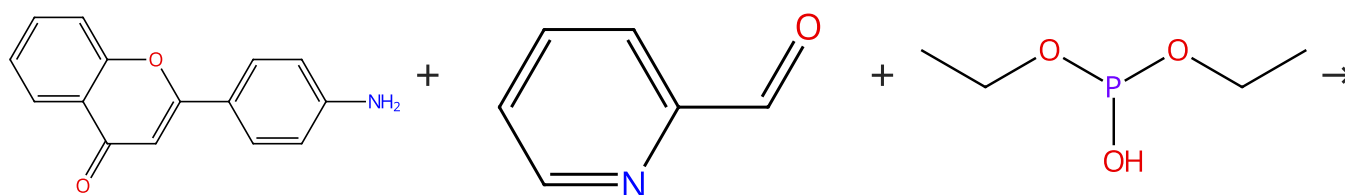
Synthetic Communications (2023), 53(23), 2041-2060.

1.1 Catalysts: Zinc oxide (ZnO); 12 min, rt

Experimental Protocols

Scheme 30 (1 Reaction)

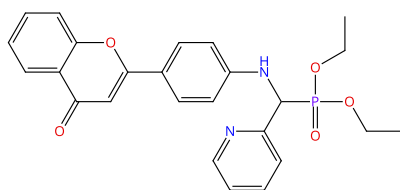
Steps: 1 Yield: 96%



Suppliers (5)

Suppliers (86)

Suppliers (10)



31-614-CAS-38142910

Steps: 1 Yield: 96%

Ultrasound mediated nano ZnO catalyzed synthesis of new α -aminophosphonates as potential anti-diabetic agents; an in silico ADMET, molecular docking study, α -amylase and α -glucosidase inhibitory activity

By: Subramanyam, CH.; et al

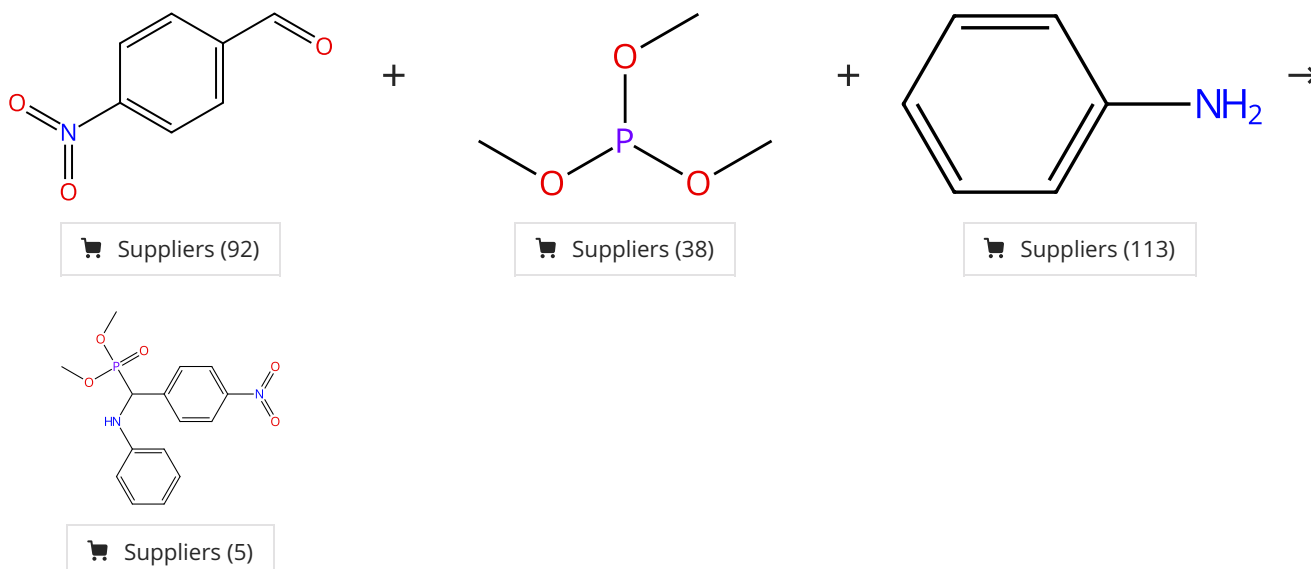
Synthetic Communications (2023), 53(23), 2041-2060.

1.1 Catalysts: Zinc oxide (ZnO); 18 min, rt

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-44718095

Steps: 1 Yield: 96%

1.1 Catalysts: Lanthanum chloride heptahydrate; 2 h, rt

Experimental Protocols

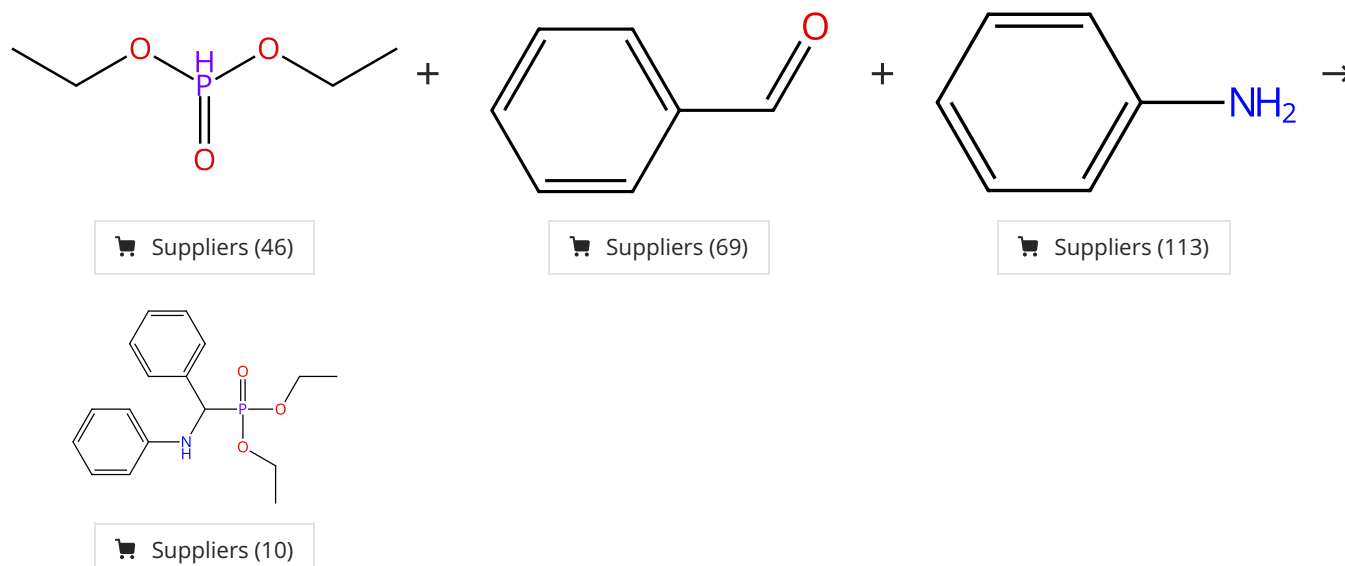
LaCl₃·7H₂O as an Effective Catalyst for the Synthesis of α -Aminophosphonates under Solvent-Free Conditions and Docking Simulation of Ligand Bond Complexes of Cyclin-Dependent Kinase 2

By: Mirzaei, Mahdi; et al

Polycyclic Aromatic Compounds (2022), 42(9), 5882-5892.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-47217966

Steps: 1 Yield: 96%

1.1 Catalysts: *N*-[1-(Hydroxymethyl)propyl]-4-nitrobenzenesulfonamide

Solvents: Ethanol; 15 min, 90 °C; 90 °C → rt

1.2 Reagents: Water; rt

Experimental Protocols

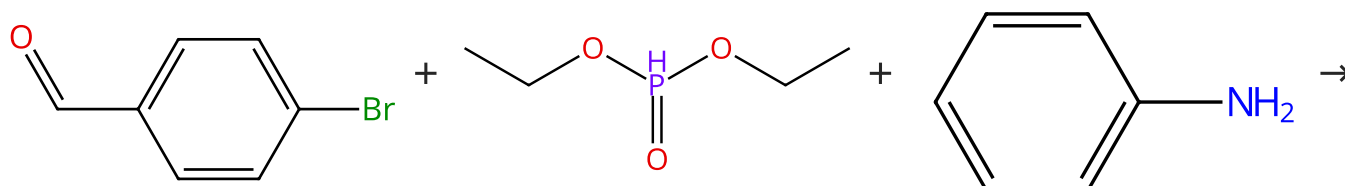
N-(1-hydroxybutan-2-yl)-4-nitrobenzene sulfonamide: A novel organocatalyst for an effective one-pot synthesis of α -amino phosphonates

By: Jaydeokar, Swati; et al

Results in Chemistry (2025), 16, 102295.

Scheme 33 (1 Reaction)

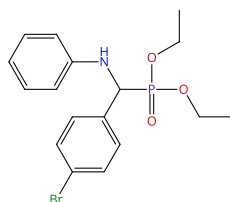
Steps: 1 Yield: 96%



Suppliers (91)

Suppliers (46)

Suppliers (113)



Suppliers (5)

31-614-CAS-47217968

Steps: 1 Yield: 96%

1.1 **Catalysts:** *N*-[1-(Hydroxymethyl)propyl]-4-nitrobenzenesulfonamide

Solvents: Ethanol; 10 min, 90 °C; 90 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

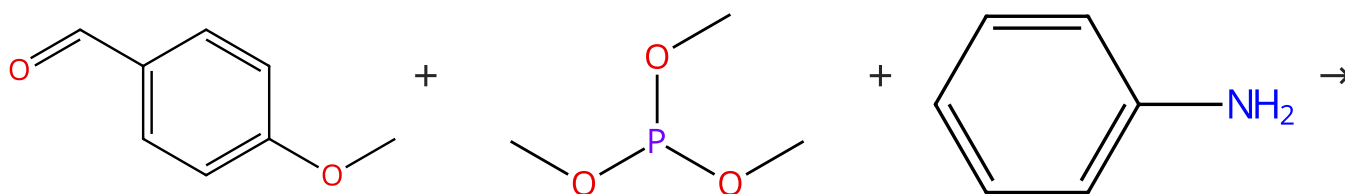
N-(1-hydroxybutan-2-yl)-4-nitrobenzene sulfonamide: A novel organocatalyst for an effective one-pot synthesis of α-amino phosphonates

By: Jaydeokar, Swati; et al

Results in Chemistry (2025), 16, 102295.

Scheme 34 (1 Reaction)

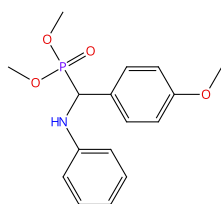
Steps: 1 Yield: 96%



Suppliers (128)

Suppliers (38)

Suppliers (113)



Suppliers (4)

31-614-CAS-44718097

Steps: 1 Yield: 96%

1.1 **Catalysts:** Lanthanum chloride heptahydrate; 0.45 h, rt

Experimental Protocols

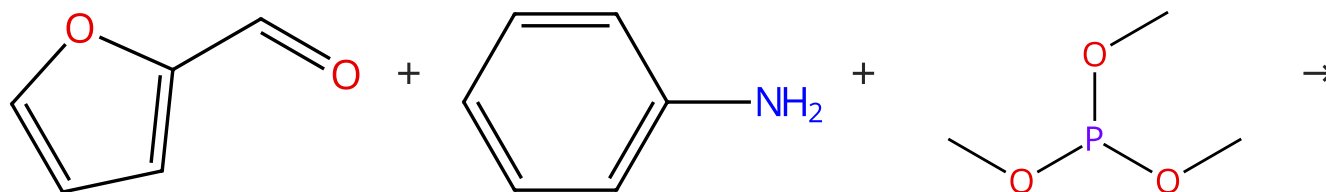
LaCl₃·7H₂O as an Effective Catalyst for the Synthesis of α-Aminophosphonates under Solvent-Free Conditions and Docking Simulation of Ligand Bond Complexes of Cyclin-Dependent Kinase 2

By: Mirzaei, Mahdi; et al

Polycyclic Aromatic Compounds (2022), 42(9), 5882-5892.

Scheme 35 (1 Reaction)

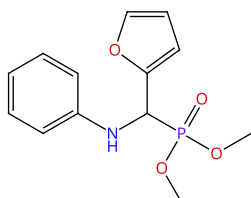
Steps: 1 Yield: 96%



Suppliers (81)

Suppliers (113)

Suppliers (38)



Suppliers (3)

31-614-CAS-29030967

Steps: 1 Yield: 96%

Coupling of aldehydes, amines, and trimethyl phosphite promoted by Amberlyst-15: highly efficient synthesis of α -aminophosphonates

1.1 Solvents: Acetonitrile; > 1 min, rt

1.2 Catalysts: Amberlyst 15; rt; 25 min, rt

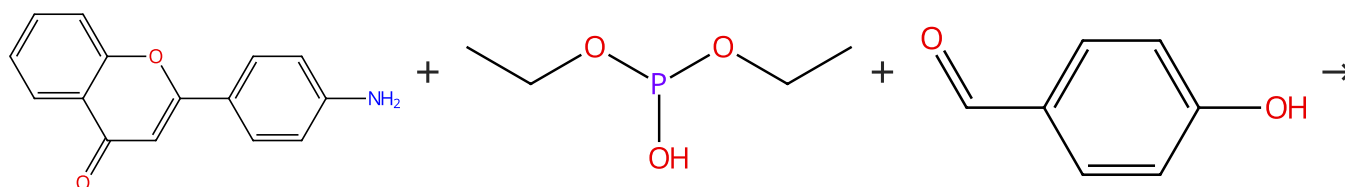
Experimental Protocols

By: Tajbakhsh, Mahmood; et al

Synthesis (2008), (3), 352-354.

Scheme 36 (1 Reaction)

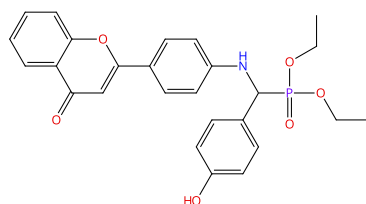
Steps: 1 Yield: 95%



Suppliers (5)

Suppliers (10)

Suppliers (121)



31-614-CAS-38142905

Steps: 1 Yield: 95%

Ultrasound mediated nano ZnO catalyzed synthesis of new α -aminophosphonates as potential anti-diabetic agents; an in silico ADMET, molecular docking study, α -amylase and α -glucosidase inhibitory activity

1.1 Catalysts: Zinc oxide (ZnO); 18 min, rt

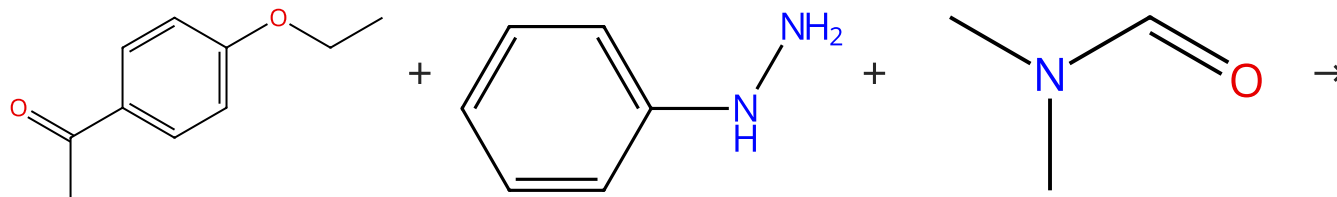
Experimental Protocols

By: Subramanyam, CH.; et al

Synthetic Communications (2023), 53(23), 2041-2060.

Scheme 37 (1 Reaction)

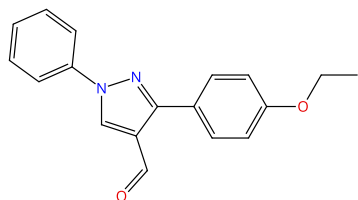
Steps: 1 Yield: 95%



Suppliers (92)

Suppliers (64)

Suppliers (304)



Suppliers (4)

31-614-CAS-28221914

Steps: 1 Yield: 95%

Highly efficient one-pot synthesis of α -aminophosphonates using nanoporous AISBA-15 catalyst in a three-component system

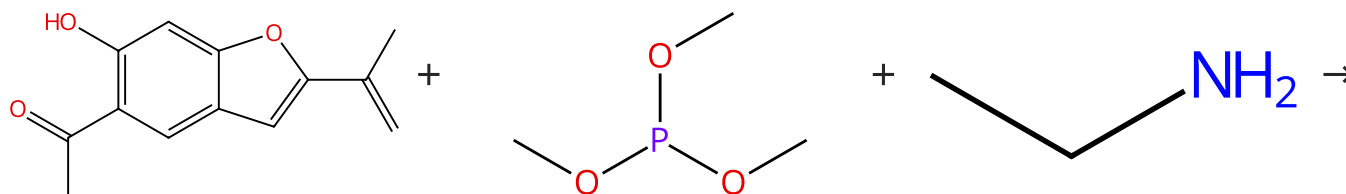
By: Kolli, Murali Krishna; et al

Research on Chemical Intermediates (2020), 46(2), 1047-1064.

- 1.1 **Catalysts:** Trifluoroacetic acid
Solvents: Dichloromethane; 6 h, rt
- 1.2 **Reagents:** Phosphorus oxychloride
Solvents: Dimethylformamide; 3 h, rt $\rightarrow$ 70 °C
- 1.3 **Reagents:** Water; cooled
- 1.4 **Reagents:** Sodium bicarbonate; pH 8, 0 °C

Scheme 38 (1 Reaction)

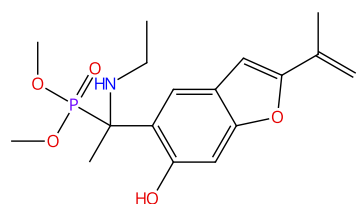
Steps: 1 Yield: 95%



Suppliers (37)

Suppliers (38)

Suppliers (86)



31-614-CAS-27978260

Steps: 1 Yield: 95%

Green synthesis of α -aminophosphonates using ZnO nanoparticles as an efficient catalyst

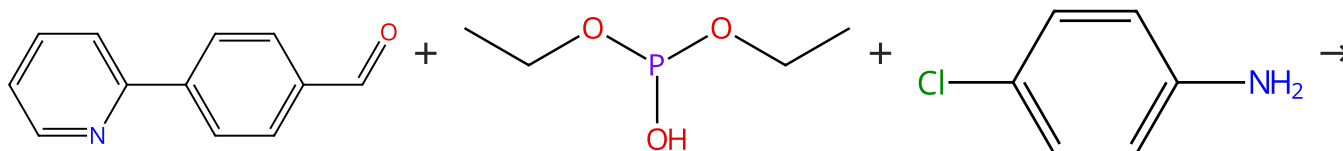
By: Ezzatzadeh, Elham

Zeitschrift fuer Naturforschung, B: A Journal of Chemical Sciences (2018), 73(3-4), 179-184.

- 1.1 **Catalysts:** Zinc oxide (ZnO)
Solvents: Water; 15 min, rt; 1 h, rt

Scheme 39 (1 Reaction)

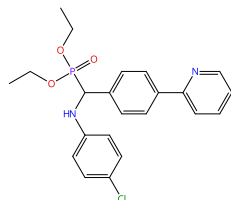
Steps: 1 Yield: 95%



Suppliers (99)

Suppliers (10)

Suppliers (98)



31-614-CAS-27579873

Steps: 1 Yield: 95%

Design, synthesis, antioxidant, and anti-breast cancer activities of novel diethyl(alkyl/aryl/heteroaryl amino)(4-(pyridin-2-yl)phenyl)methylphosphonates

By: Prasad, Sthanikam Siva; et al

Archiv der Pharmazie (Weinheim, Germany) (2013), 346(5), 380-391.

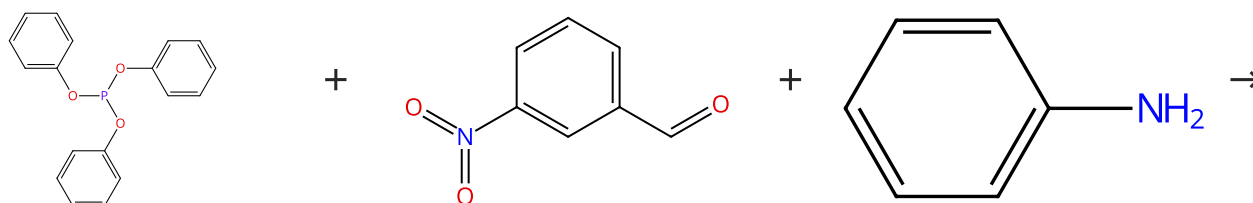
1.1 Catalysts: Cupric acetate; 5.5 min

1.2 Reagents: Water

Experimental Protocols

Scheme 40 (1 Reaction)

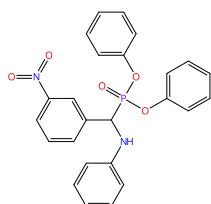
Steps: 1 Yield: 95%



Suppliers (63)

Suppliers (90)

Suppliers (113)



31-614-CAS-37833274

Steps: 1 Yield: 95%

Pvp-based deep eutectic solvent polymer: Sustainable Bronsted-Lewis acidic catalyst in the synthesis of α -aminophosphonate and bisindole

By: Shahiri Haghayegh, Mehran; et al

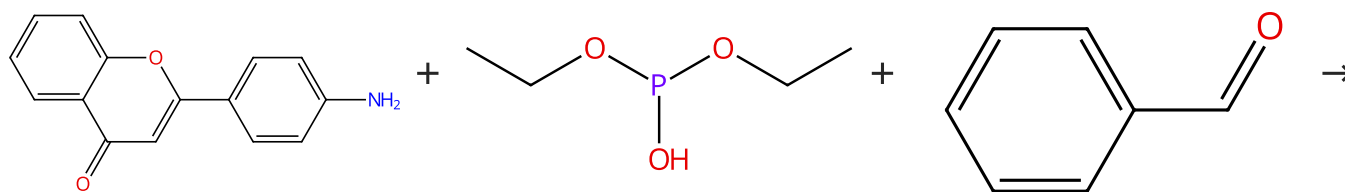
Journal of Molecular Liquids (2023), 387, 122677.

1.1 Catalysts: Zinc chloride, Poly(vinylpyrrolidone) (quaternized)
Solvents: Ethanol, Water; 90 min, rt

Experimental Protocols

Scheme 41 (1 Reaction)

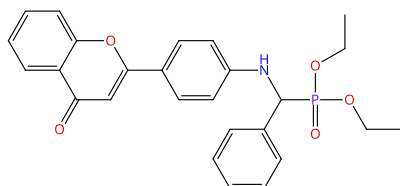
Steps: 1 Yield: 95%



Suppliers (5)

Suppliers (10)

Suppliers (69)



31-614-CAS-38142903

Steps: 1 Yield: 95%

Ultrasound mediated nano ZnO catalyzed synthesis of new α -aminophosphonates as potential anti-diabetic agents; an in silico ADMET, molecular docking study, α -amylase and α -glucosidase inhibitory activity

By: Subramanyam, CH.; et al

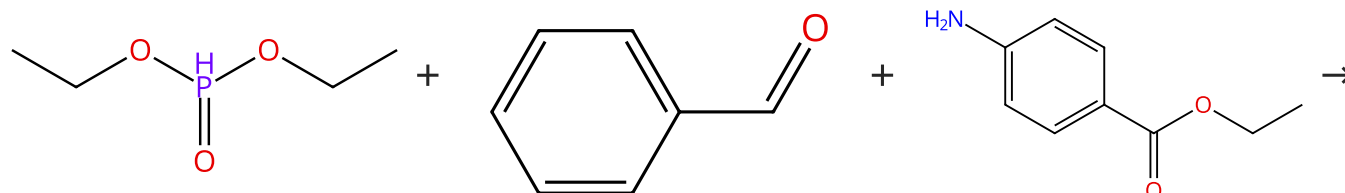
Synthetic Communications (2023), 53(23), 2041-2060.

1.1 Catalysts: Zinc oxide (ZnO); 15 min, rt

Experimental Protocols

Scheme 42 (1 Reaction)

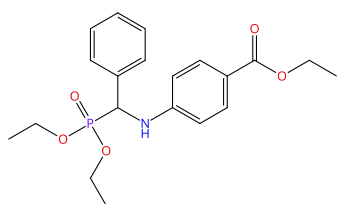
Steps: 1 Yield: 95%



Suppliers (46)

Suppliers (69)

Suppliers (127)



31-614-CAS-47217991

Steps: 1 Yield: 95%

N-(1-hydroxybutan-2-yl)-4-nitrobenzene sulfonamide: A novel organocatalyst for an effective one-pot synthesis of α -amino phosphonates

By: Jaydeokar, Swati; et al

Results in Chemistry (2025), 16, 102295.

1.1 Catalysts: N-[1-(Hydroxymethyl)propyl]-4-nitrobenzenesulfonamide

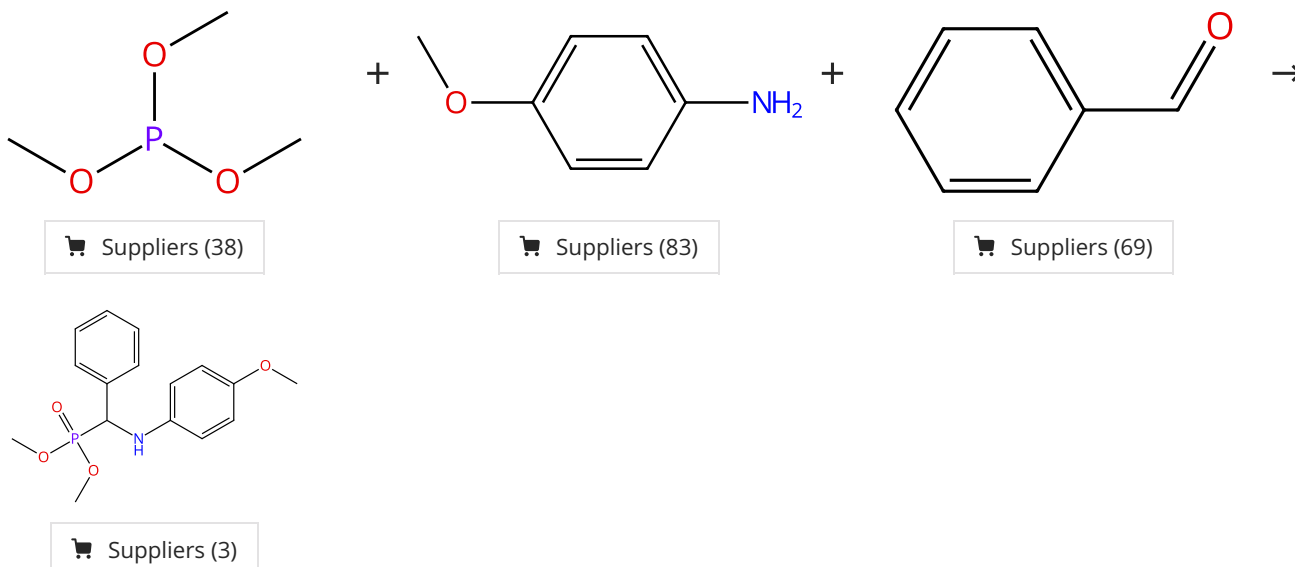
Solvents: Ethanol; 10 min, 90 °C; 90 °C → rt

1.2 Reagents: Water; rt

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 95%



31-614-CAS-44718101

Steps: 1 Yield: 95%

LaCl₃·7H₂O as an Effective Catalyst for the Synthesis of α-Aminophosphonates under Solvent-Free Conditions and Docking Simulation of Ligand Bond Complexes of Cyclin-Dependent Kinase 2

By: Mirzaei, Mahdi; et al

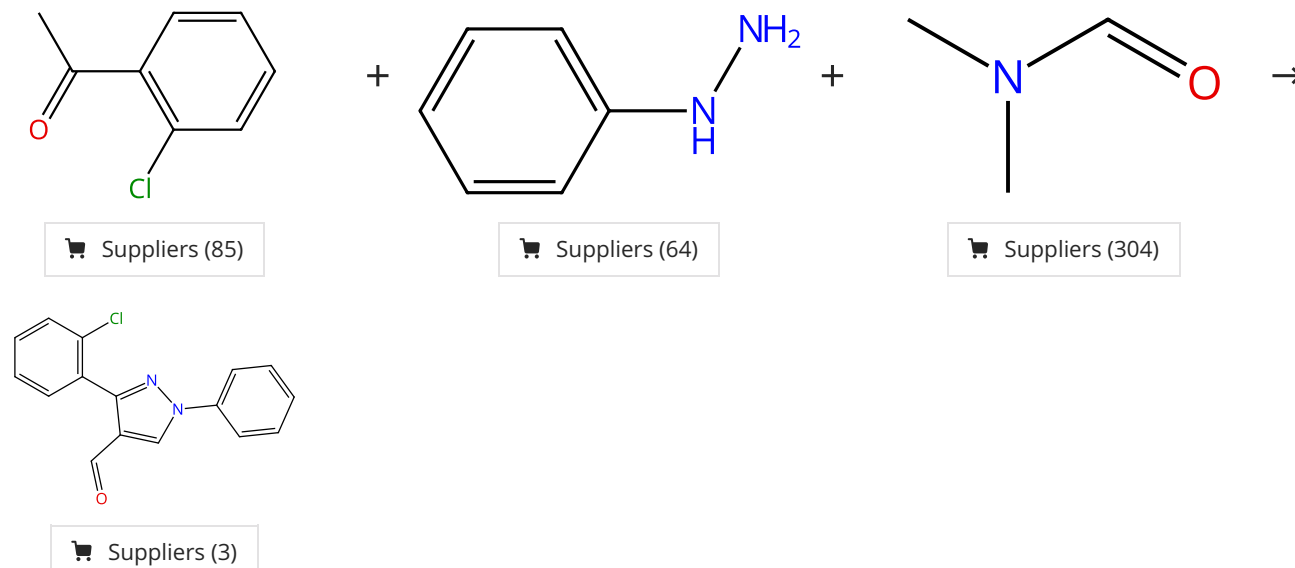
Polycyclic Aromatic Compounds (2022), 42(9), 5882-5892.

1.1 Catalysts: Lanthanum chloride heptahydrate; 2 h, rt

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 95%



31-614-CAS-26622819

Steps: 1 Yield: 95%

Highly efficient one-pot synthesis of α-aminophosphonates using nanoporous AISBA-15 catalyst in a three-component system

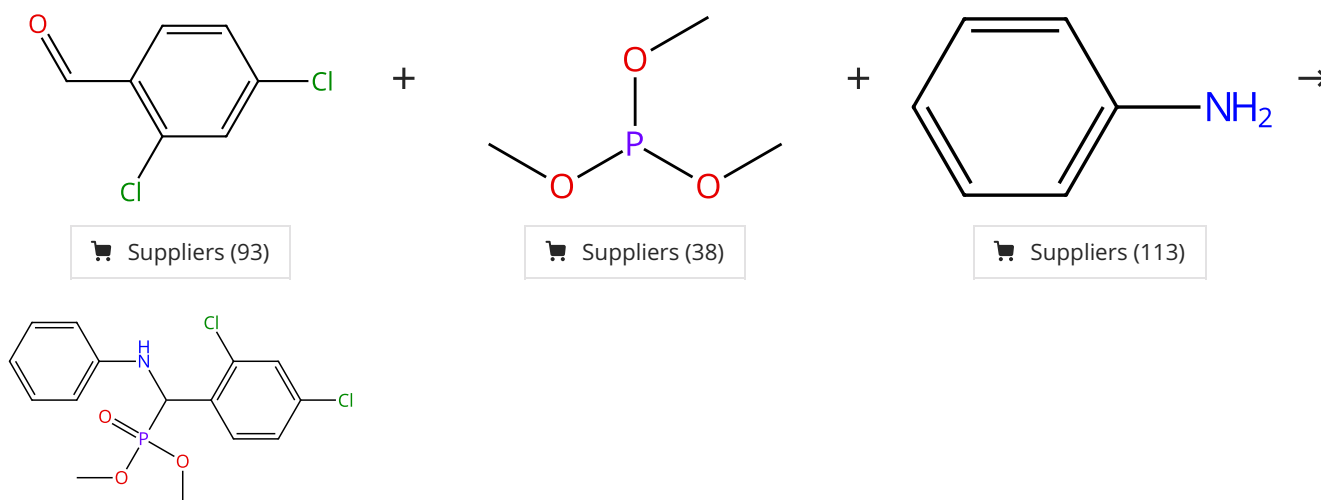
By: Kolli, Murali Krishna; et al

Research on Chemical Intermediates (2020), 46(2), 1047-1064.

- 1.1 Catalysts: Trifluoroacetic acid
Solvents: Dichloromethane; 6 h, rt
- 1.2 Reagents: Phosphorus oxychloride
Solvents: Dimethylformamide; 3 h, rt → 70 °C
- 1.3 Reagents: Water; cooled
- 1.4 Reagents: Sodium bicarbonate; pH 8, 0 °C

Scheme 45 (1 Reaction)

Steps: 1 Yield: 95%



31-614-CAS-24720828

Steps: 1 Yield: 95%

- 1.1 **Catalysts:** Iron oxide (Fe₃O₄) (nanocomposite with phosphotungstic acid and carbon nitride), 12-Tungstophosphoric acid (nanocomposite with triiron tetroxide and carbon nitride), Carbon nitride (C₃N₄) (nanocomposite with phosphotungstic acid and triiron tetroxide)
Solvents: Ethanol; 1 - 3 h, 25 °C

Experimental Protocols

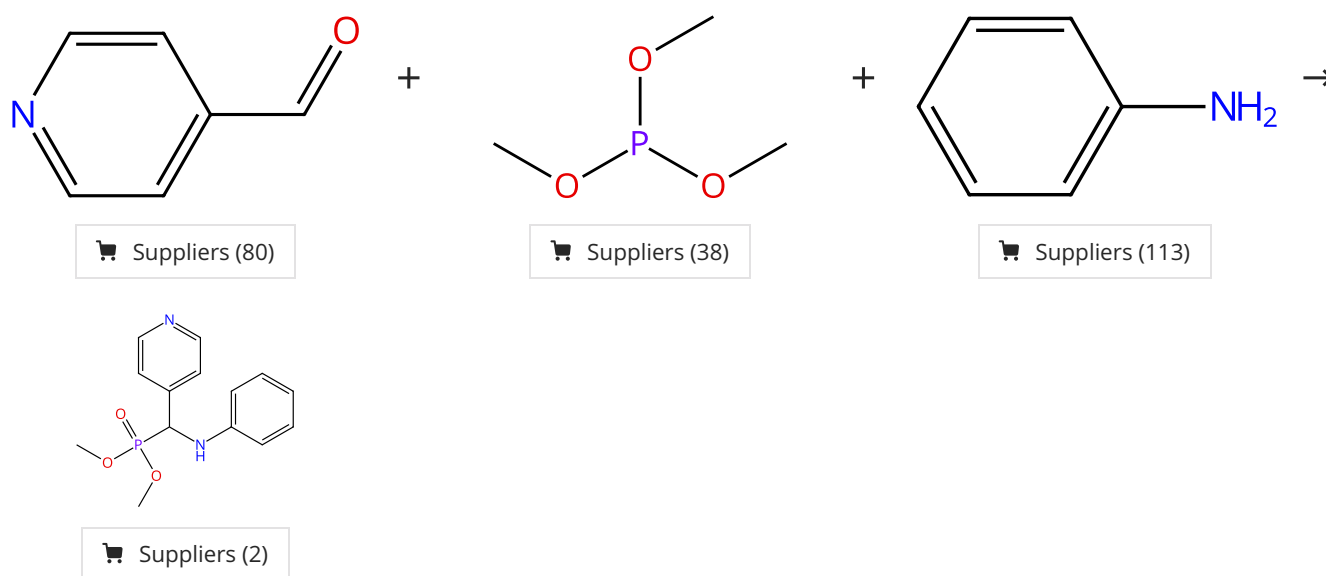
Design and preparation of HPW-anchored magnetic carbon nitride nanosheets: an efficient and eco-friendly nanocomposite for one-pot synthesis of α-amino phosphonates

By: Azhdari, Asieh; et al

Research on Chemical Intermediates (2021), 47(12), 4915-4928.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 95%



31-614-CAS-37833256

Steps: 1 Yield: 95%

- 1.1 **Catalysts:** Zinc chloride, Poly(vinylpyrrolidone) (quaternized)
Solvents: Ethanol, Water; 15 min, rt

Experimental Protocols

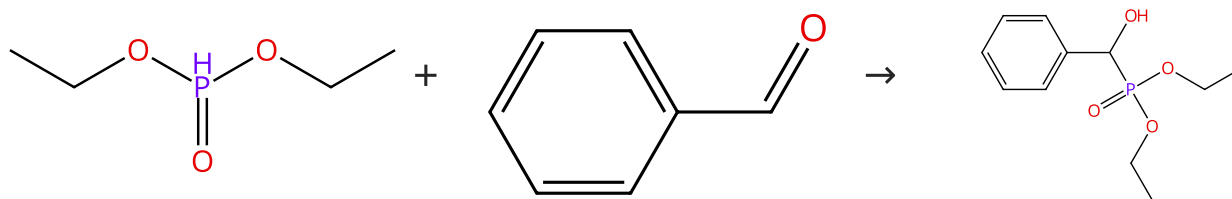
Pvp-based deep eutectic solvent polymer: Sustainable Bronsted-Lewis acidic catalyst in the synthesis of α-amino phosphonate and bisindole

By: Shahiri Haghayegh, Mehran; et al

Journal of Molecular Liquids (2023), 387, 122677.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (46)

Suppliers (69)

Suppliers (32)

31-614-CAS-35739714

Steps: 1 Yield: 95%

Optimized Synthesis and Cytotoxic Activity of α -Aminophosphonates Against a Multidrug Resistant Uterine Sarcoma Cell Line

By: Varga, Petra Regina; et al

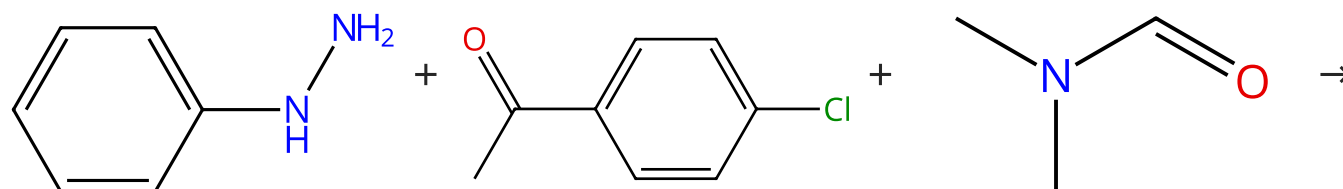
Letters in Drug Design & Discovery (2023), 20(3), 365-371.

1.1 **Reagents:** Triethylamine
Solvents: Acetone; 2.5 h, reflux

Experimental Protocols

Scheme 48 (1 Reaction)

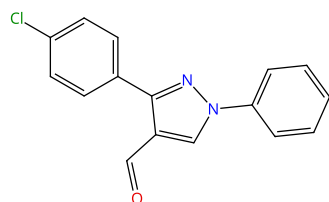
Steps: 1 Yield: 95%



Suppliers (64)

Suppliers (67)

Suppliers (304)



Suppliers (58)

31-614-CAS-27308384

Steps: 1 Yield: 95%

Highly efficient one-pot synthesis of α -aminophosphonates using nanoporous AISBA-15 catalyst in a three-component system

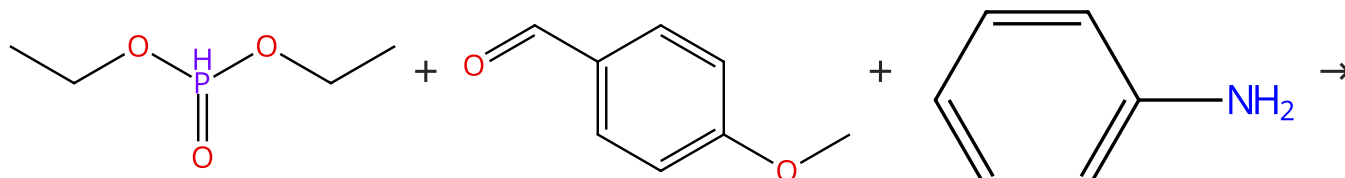
By: Kolli, Murali Krishna; et al

Research on Chemical Intermediates (2020), 46(2), 1047-1064.

- 1.1 **Catalysts:** Trifluoroacetic acid
Solvents: Dichloromethane; 6 h, rt
- 1.2 **Reagents:** Phosphorus oxychloride
Solvents: Dimethylformamide; 3 h, rt $\rightarrow$ 70 °C
- 1.3 **Reagents:** Water; cooled
- 1.4 **Reagents:** Sodium bicarbonate; pH 8, 0 °C

Scheme 49 (1 Reaction)

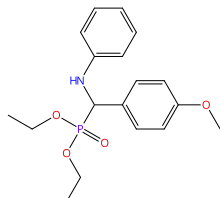
Steps: 1 Yield: 95%



Suppliers (46)

Suppliers (128)

Suppliers (113)



Suppliers (4)

31-614-CAS-47217977

Steps: 1 Yield: 95%

1.1 **Catalysts:** *N*-[1-(Hydroxymethyl)propyl]-4-nitrobenzenesulfonamide

Solvents: Ethanol; 10 min, 90 °C; 90 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

N-(1-hydroxybutan-2-yl)-4-nitrobenzene sulfonamide: A novel organocatalyst for an effective one-pot synthesis of α-amino phosphonates

By: Jaydeokar, Swati; et al

Results in Chemistry (2025), 16, 102295.